NTU VeNTURe Programme Structure

C22 Pte Ltd — veNTURe July 2026 Cohort

1. Programme Overview

Contrarian Market Strategy & Behavioural Systems Design (Data Analytics & Strategy)

An 12-week industry challenge where NTU students work in teams to solve real-world problems using AI, finance, data science and technology. Students will receive mentorship, industry exposure and compete for prizes based on their final solution.

Key Information

Host Company: C22 Pte Ltd

Duration: 12 weeks (core build 10 July – 10 October; Awards 20 October)

Selection & Confirmation: 1-9 July

Programme Structure

The challenge runs in six progressive stages. Data access and accountability increase at each stage.

Stage 1	**Data Exploration**	Learn the dataset, run exploratory analysis, generate hypotheses and submit first insights.
Stage 2	**Feature Engineering**	Engineer behavioral features — risk-taking metrics, consistency indicators, behavioral sequences and emotional-trading signals.
Stage 3	**Model Building**	Build prediction models on historical data, assessed on accuracy, robustness, explainability and novelty.
Stage 4	**Live Validation**	Selected models are evaluated by C22 against new incoming data: Historical → Hidden Validation → Live Evaluation.
Stage 5	**Research Defence**	Finalists explain why their model works, its failure cases, how it adapts, and the improvements they would make.
Stage 6	**Production Candidate**	Outstanding models may progress toward production research with C22 beyond the programme.

2. Challenge Statement

Financial markets are driven not only by price movements, but by predictable patterns in human behaviour. Many participants exhibit consistent decision-making biases that lead to suboptimal outcomes.

The company is exploring how to systematically identify and leverage these behavioural inefficiencies to build robust, data-driven strategies. At the same time, we are interested in how different system structures AKA game rules influence participant behaviour and overall outcomes.

Tasks:

- Analyze trading datasets to uncover behavioral patterns and inefficiencies.
- Develop and test contrarian or alternative strategy frameworks.
- Explore how rule design (constraints, incentives, structure) impacts participant decisions.
- Build and simulate models to evaluate performance under different scenarios.
- Present insights and strategic recommendations.

3. Deliverables

Deliverables are staged. Each stage produces a checkpoint submission that builds toward the three final graded deliverables, so C22 can track progress and award standing throughout — not only at the end.

Stage Checkpoints (contribute to cumulative standing)

Stage 1 · Data Exploration: EDA summary and hypotheses memo.

Stage 2 · Feature Engineering: Documented behavioural feature set (code plus a short brief).

Stage 3 · Model Building: Trained model with historical backtest and validation results.

Stage 4 · Live Validation: Frozen model submitted for C22 hidden and live evaluation + short performance note.

Stage 5 · Research Defence: Final pitch and defence Q&A.

Final Deliverables (graded)

- 1 Final presentation (key insights + metrics + strategic recommendations)
- 1 Written report (analysis, models explored, conclusions)
- 1 Strategy framework or prototype (e.g. model logic, rule system, or simulation)

4. Mentorship Structure & Resources Provided

Bi-monthly check-ins, at the request of solos/teams. Selected dates will be updated to all in advance.

Resources provided: anonymized trading datasets, a glossary of domain terms, a pre-start briefing

5. Judging Criteria & Prize Structure

Technical Quality – 30% (rigour of analysis, modelling and validation)
Business Value – 30% (relevance and impact of insights for C22)
Practical Implementation – 30% (feasibility and robustness on unseen data)
Innovation – 10% (originality of features and overall approach)

3 prizes for 3 winners, in descending order – S$2000, S$1500, S$1000.

6. Rules & Programme Flow

IP ownership by default will lie with the company, as we will be providing proprietary data to work on.

Confidentiality, and academic integrity, including PDPA agreement needed to handle sensitive data.

Programme Flow (12 Weeks)

Briefing & Problem Release	10 July
Stage 1 · Data Exploration	10 – 24 July (Week 1-2)
Stage 2 · Feature Engineering	25 July – 7 August (Weeks 3-4)
Stage 3 · Model Building	8 – 21 August (Week 5-6)
Midpoint Review	22 – 28 August
Stage 4 · Live Validation	29 August – 11 September (Weeks 7–8)
Iteration & Refinement	11 – 25 September (Week 9–10)
Stage 5 · Research Defence & Final Submission	26 September – 9 October (Week 11–12)
Judging	10 October
Awards	20 October